Template Screening Report (v5.7.1 - Multi-Source AI Evaluator)

Journal Template: Jurnal Teknologi

Submission No: 26799

Author Name: Surbakti et al.

Article Title: VERIFICATION OF SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER 35 YEARS OF OPERATIONS

Source File: JT-tukiran-utm2026.docx

Screened PDF: JT-tukiran-utm2026.pdf

Screening Date: 2026-05-21 09:40:04

Detection Source Debug Report

Item	DOC Detection	PDF Detection	AI Detection	Final Source
Article Title	NO	YES	YES	PDF + AI
Authors	NO	YES	YES	PDF + AI
Affiliations	NO	YES	YES	PDF + AI
Corresponding Author	YES	YES	SKIPPED	DOC + PDF
Graphical Abstract	YES	YES	SKIPPED	DOC + PDF
English Abstract	YES	YES	SKIPPED	DOC + PDF
Keywords	YES	YES	SKIPPED	DOC + PDF
Malay Abstract (Abstrak)	NO	YES	YES	PDF + AI
Kata Kunci	NO	NO	NO	None
Introduction	YES	NO	YES	DOC + AI
Methodology	YES	NO	YES	DOC + AI
Results and Discussion	YES	YES	SKIPPED	DOC + PDF
Conclusion	YES	NO	YES	DOC + AI
Acknowledgement	YES	YES	SKIPPED	DOC + PDF
References	YES	YES	SKIPPED	DOC + PDF

Summary: DOC detected 10 items | PDF detected 11 items | AI confirmed 7 items

The system uses DOC + PDF + AI voting logic. AI serves as a third checker when DOC and PDF conflict. AI is skipped when DOC and PDF already agree.

Executive Summary

Overall Decision: NEEDS MANUAL REVIEW

Main Reason: The manuscript generally follows the template but contains inconsistencies that require manual review.

Critical Failed Items: None

Items Needing Review:

• Article Title

• Authors

• Affiliations

• Malay Abstract (Abstrak)

• Introduction

Suggested Action: Editor should review the highlighted issues before proceeding.

Overall Result

Metric	Score / Value
Final Status	NEEDS MANUAL REVIEW
Final Score	76/100
Structure Score	47%
Layout Score	89%
Formatting Score	100%
Detection Method	DOCX-First + Multi-Source AI (v5.7.1)
Decision	MANUAL REVIEW

How Items Are Evaluated

DOC + PDF + AI SKIPPED	PASS	Both deterministic detectors agree
DOC + PDF + AI CONFIRMED	PASS	All three confirm
DOC + AI	NEEDS REVIEW	DOC+AI confirm, but PDF layout conflicts
PDF + AI	NEEDS REVIEW	PDF+AI confirm, but DOC source extraction fails
DOC only	NEEDS REVIEW	DOC detects, but PDF+AI don't confirm
PDF only	NEEDS REVIEW	PDF detects, but DOC+AI don't confirm
AI only	NEEDS REVIEW	Only AI confirms (deterministic detectors failed)
None	FAILED	All three fail to detect required item

Status Explanation

PASS — Item follows the template structure, position, and formatting.

WARNING/NEEDS REVIEW — Item exists but evidence is inconsistent or layout differs from template.

FAILED — Required item truly missing after DOC/PDF/AI checks.

Item Screening Details

Item	DOC	PDF	AI	Final Source	Status	Reason
Article Title	NO	YES	YES	PDF + AI	WARNING	Title detected by PDF and AI, but DOC source text did not co
Authors	NO	YES	YES	PDF + AI	WARNING	Authors detected by PDF and AI, but DOC source text did not
Affiliations	NO	YES	YES	PDF + AI	WARNING	Affiliations detected by PDF and AI, but DOC source text did
Corresponding Author	YES	YES	SKIPPED	DOC + PDF	PASS	Corresponding_Author confirmed by both DOC and PDF detection

Graphical Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Graphical_Abstract confirmed by both DOC and PDF detection.
English Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Abstract confirmed by both DOC and PDF detection.
Keywords	YES	YES	SKIPPED	DOC + PDF	PASS	Keywords confirmed by both DOC and PDF detection.
Malay Abstract (Abstrak)	NO	YES	YES	PDF + AI	WARNING	Abstrak detected by PDF and AI, but DOC source text did not
Kata Kunci	NO	NO	NO	None	FAILED	Kata_Kunci not found by DOC, PDF, or AI. [AI: weak_evidence,
Introduction	YES	NO	YES	DOC + AI	WARNING	Introduction confirmed by DOC and AI, but PDF/layout detecti
Methodology	YES	NO	YES	DOC + AI	WARNING	Methodology confirmed by DOC and AI, but PDF/layout detectio
Results and Discussion	YES	YES	SKIPPED	DOC + PDF	PASS	Results_Discussion confirmed by both DOC and PDF detection.
Conclusion	YES	NO	YES	DOC + AI	WARNING	Conclusion confirmed by DOC and AI, but PDF/layout detection
Acknowledgement	YES	YES	SKIPPED	DOC + PDF	PASS	Acknowledgement confirmed by both DOC and PDF detection.
References	YES	YES	SKIPPED	DOC + PDF	PASS	References confirmed by both DOC and PDF detection.

Detailed Findings

Article Title [title]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Article Title' section must exist.
Found: Matched heading: "VERIFICATION SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER " | Detection method: score_rank
Confidence: 1.00
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 1.00
AI Evidence: PDF snippet shows the full title and author information clearly, while DOC snippet does not contain the title section at all.
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Jurnal Teknologi, Full Paper, Article history
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, top area
Found: Page 1, left column, upper area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.10).

C. Formatting Evaluation

Expected: Heading should be in uppercase style.
Expected: Heading should use bold font.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Authors [authors]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Authors' section must exist.
Found: Matched heading: "VERIFICATION SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER " | Detection method: score_rank
Confidence: 0.70
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.85
AI Evidence: PDF snippet shows a full authors section with multiple authors and affiliations; DOC snippet does not contain any authors section, only other manuscri
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Article history, Received, Accepted
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left column, upper area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Affiliations [affiliations]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Affiliations' section must exist.
Found: Matched heading: "Nuclear Energy, National Research and Innovation Agency (BRIN), KST BJ Habiebi, Serpong Gd. 80, Tan" | Detection method: score_rank
Confidence: 0.70
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.85
AI Evidence: PDF snippet shows full Affiliations section with author names and institutional addresses. DOC snippet does not contain the Affiliations section, only
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Accepted
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left column, upper area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Corresponding Author [corresponding_author]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Corresponding Author' section must exist.
Found: Matched heading: "Corresponding author" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Corresponding Author' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, right-center area, upper area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS

Reason: Formatting matches template expectations.
Overall Item Status: PASS

Graphical Abstract [graphical_abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Graphical Abstract' component must exist.
Found: Matched heading: "Graphical abstract" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Graphical Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, upper area
Found: Page 1, left-center area, upper-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Image should be present.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

English Abstract [abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'English Abstract' section must exist.
Found: Matched heading: "Abstract" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'English Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left-center area, upper-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

Keywords [keywords]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Keywords' section must exist.
Found: Matched heading: "Keywords" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Keywords' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, lower-middle area
Found: Page 1, left-center area, lower-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

Malay Abstract (Abstrak) [abstrak]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Malay Abstract (Abstrak)' section must exist.
Found: Nearby content: "78:1 (2016) 1–5 | www.jurnalteknologi.utm.my | eISSN 2180–3722 | Jurnal Teknologi Full Paper VERIFICATION SHUTDOWN MARGIN AND EXC
Confidence: 0.00
Detection Method: semantic_fallback
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.70
AI Evidence: PDF snippet shows manuscript metadata and partial content consistent with presence of Abstrak; DOC snippet is missing this content entirely.
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Kata Kunci [kata_kunci]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Kata Kunci' section must exist.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO

PDF Detection: NO

AI Detection: NO

AI Evidence: DOC snippet includes title, authors, affiliations, and article history but no keywords. PDF snippet similarly includes title, authors, affiliations, a

Evidence Sources: None

Final Source Used: None

Status: FAILED

Reason: Required component 'Kata Kunci' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: FAILED

Introduction [introduction]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Introduction' section must exist.

Found: Matched heading: "1.0. INTRODUCTION" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: NO

AI Detection: YES

AI Confidence: 0.85

AI Evidence: DOC snippet provides detailed introduction content about the reactor, regulatory context, and study objectives. PDF snippet lacks Introduction content

Evidence Sources: DOC + AI

Final Source Used: DOC + AI

Rejected false positives: 1.0. INTRODUCTION

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area

Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Methodology [methodology]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Methodology' section must exist.

Found: Matched heading: "3.0. METHODOLOGY" | Detection method: source_docx_text

Confidence: 0.85

Detection Method: source_docx_text

DOC Detection: YES

PDF Detection: NO

AI Detection: YES

AI Confidence: 0.85

AI Evidence: DOC snippet includes detailed methodology content about core calculations and modeling using the WIMSD-5B program. PDF snippet only contains header an

Evidence Sources: DOC + AI
Final Source Used: DOC + AI
Rejected false positives: 3.0. METHODOLOGY
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Results and Discussion [results_discussion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Results and Discussion' section must exist.
Found: Matched heading: "4.0. RESULTS AND DISCUSSIONS" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Results and Discussion' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, center area, lower area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

Conclusion [conclusion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Conclusion' section must exist.
Found: Matched heading: "4.0. CONCLUSION" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: YES
AI Confidence: 0.85
AI Evidence: DOC snippet includes a detailed Conclusion discussing verification calculations, safety parameters, and operational limits. PDF snippet lacks any Conc
Evidence Sources: DOC + AI
Final Source Used: DOC + AI
Rejected false positives: 4.0. CONCLUSION
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Acknowledgement [acknowledgement]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Acknowledgement' component is recommended.
Found: Matched heading: "Acknowledgment" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Acknowledgement' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, left-center area, upper-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

References [references]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'References' section must exist.
Found: Matched heading: "References" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'References' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, left column, lower-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: PASS

Editor Conclusion

The manuscript generally follows the template but has items that require manual review. These items had inconsistent evidence between DOC detection, PDF layout analysis, or AI verification. Please review the highlighted items before making a final decision.

Structure Score: 47% (Weight: 40%)

Layout Score: 89% (Weight: 30%)

Formatting Score: 100% (Weight: 30%)

Final Score: 76/100

Items Needing Review:

• Article Title — WARNING
• Authors — WARNING
• Affiliations — WARNING
• Malay Abstract (Abstrak) — WARNING
• Introduction — WARNING

This report was automatically generated by Turnitin Assistant v5.7.1 Template Screening Engine (DOCX-First + Multi-Source AI Evaluator).

Jurnal
Teknologi

Full Paper

VERIFICATION SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER 35 YEARS OF OPERATIONS

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XB Month 202Y

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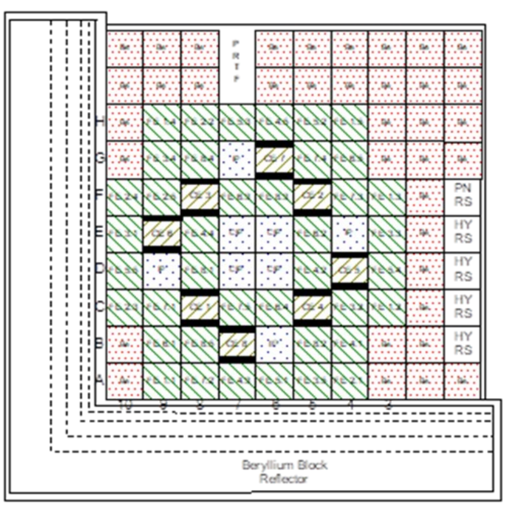
Graphical abstract



Abstract

VERIFICATION OF SHUTDOWN MARGIN AND EXCESS REACTIVITY VALUES ON GAS-RSG EQUILIBRIUM CORE AFTER 35 YEARS OF OPERATIONS. To ensure the safety of RSG-GAS reactor operation, a review of the safety parameters of core operation must be carried out, as documented in the updated Safety Analysis Report (SAR) and Periodic Safety Review (PSR) documents. This research aims to verify the value of safety-related parameters in the SAR (Safety Analysis Report), including neutronic aspects. In this aspect, verification of the reactivity is more balanced than that of the RSG-GAS after 35 years of operation through calculations using the WIMSD-5B / Batan-FUEL codes. From the calculation results obtained, the suthdown margin reactivity is 4.59%, and from experimental results is 4.10%. The deviation of both results is 12% after 35 years of operation. Meanwhile, the excess reactivity parameter on the equilibrium RSG-GAS without xenon of 7.85%. While the excess reactivity value of the RSG-GAS core experiment is 7.11%. From the two values obtained, the calculated value is in accordance with the experimental value, with a difference of 10.4%.

Keywords: *Verification, Excess Reactivity, RSG-GAS Reactor, WIMSD-5B, Batan- FUEL*

1.0. INTRODUCTION

Currently, the Multipurpose Reactor (RSG-GAS), the National Research and Innovation Agency (BRIN), as the Nuclear Installation Operator (NIO) operating the RSG-GAS reactor, will submit a revised Safety Analysis Report (SAR) document to the Nuclear Energy Regulatory Agency (BAPETEN) regarding the reactor's operating permit. In line with the revision process, BAPETEN, as a technical support unit, is conducting an independent assessment of the RSG-GAS reactor's safety related to its supervisory duties. The results of this study are used to provide technical support to the Licensing Directorate in the SAR evaluation process. This study presents the calculation of the RSG-GAS equilibrium core excess reactivity at the Beginning of Cycle (BOC) condition. This is important to do because the BOC equilibrium core excess reactivity value is a parameter directly related to the core configuration and the control rod's ability to shut down reactor operations[1,2,3]. Therefore, the initial equilibrium core excess reactivity limit is closely related to the reactor's operational safety. The calculation of the BOC equilibrium core excess reactivity was carried out using the WIMSD-5B/Batan-FUEL codes[4,5]. This study aims to obtain the excess reactivity value at the initial equilibrium core without xenon using the WIMSD-5B/Batan-FUELcode using ENDFBVIII.0 nuclear data[6.7.8]. The study was conducted to support the evaluation of the SAR and PSR (Periodic Safety Review) documents for the RSG-GAS reactor, specifically verifying the excess reactivity value at the equilibrium core (BOC). The calculation results will be compared with the experimental results of the RSG-GAS working core. This study was conducted as part of the review of the safety analysis report and periodic safety documents for the RSG-GAS core.

2.0. RSG-GAS CORE DESCRIPTION

The G.A. Siwabessy Multipurpose Reactor (RSG-GAS) is the world's first MTR (Material Testing Reactor)- type research reactor to be operated directly with low-enriched uranium (LEU) fuel elements [9,10]. At the time the RSG-GAS design was carried out, only oxide-type LEU fuel elements (U_3O_8-Al) were available, as shown in Figure 1. Therefore, RSG-GAS initially used oxide fuel with a Uranium density in the meat of 2.96 g/cm^3 with a ^{235}U enrichment of 19.75% [11]. In order to improve reactor performance, the RSG-GAS core has been converted from uranium oxide fuel to uranium silicide. This was done because the use of silicide fuel with a higher Uranium density can inc- type research reactor to be operated directly with low-enriched uranium (LEU) fuel elementsrease the length of the reactor's operating cycle [12,13]. The equilibrium core (TWC) arrangement is a configuration of the equilibrium core (TWC). The active core of the TWC consists of 40 standard fuel elements (FB), 8 control fuel elements, one large central irradiation position (CIP) consisting of 2 x 2 core grid positions, and 4 irradiation positions inside the reactor core, each taking one core grid position, so that the entire equilibrium core consists of 960 fuel element plates, which means it is identical to 45.7 standard fuel elements [14, 15].

a. Reactor Core

The reactor core consists of grids that form a 10 x 10 matrix. In the active core, the core grid will form an 8 x 8 matrix that is used as a place for fuel elements, control elements, reflector elements, and irradiation facilities. At one corner, the core is covered by an L-shaped reflector block made of beryllium, which functions as a reflector or reflector to reflect neutrons that escape from the core [16,17]. Light water (H_2O) as a coolant flows through the core components to absorb
heat and is discharged through a secondary cooling system.

b. Fuel element

The RSG-GAS design uses MTR-type U_3O_8Al fuel elements with a uranium (^{235}U) enrichment of 19.75% and a density of 2.96 g/cm^3. Each fuel element consists of 21 plates measuring 625 mm long and 70.75 mm wide, while the active fuel plate measures 1.30 mm thick, 600 mm long, and 62.75 mm wide. The plates used are made of AlMg. Each fuel element contains an average of 250 grams of uranium. However, currently, the RSG-GAS uses 250 g of silicide fuel with the same geometry and other parameters as the design fuel [8].

Under typical working core conditions (TWC), there are 40 fuel elements and a burn-up distribution (burn fraction) consisting of 8 groups with an average initial burn-up of 23.3% and an average final burn-up of 31.3%. In each cycle, five fuel elements and one control element are replaced, with an average burn-in fraction of approximately 56% [18, 19].

c. Control Elements

The control rods used in the RSG-GAS are made of AgInCd (85%, 10%, and 5%) in a fork-shaped structure and are wrapped in stainless steel [20]. Each control element consists of 15 fuel element plates and two neutron absorber blades. During operation, the RSG-GAS uses eight control rods, consisting of seven shim rods and one regulating rod. These control rods are placed at specific positions within the reactor core. The position of the control rods within the core can be seen in Figure 1.

The RSG-GAS design uses MTR-type U3O8Al fuel elements with a uranium (235U) enrichment of 19.75% and a density of 2.96 g/cm^3. Each fuel element consists of 21 plates measuring 625 mm long and 70.75 mm wide, while the active fuel plate measures 1.30 mm thick, 600 mm long, and 62.75 mm wide. The plates used are made of AlMg. Each fuel element contains an average of 250 grams of uranium. However, currently, the RSG-GAS uses 250 g of silicide fuel with the same geometry and other parameters as the design fuel [21].

3.0. METHODOLOGY

Calculations were performed on a balanced core with a nominal power of 15 MW and free of xenon and samarium. The calculation steps are as follows:

3.1. Cell Calculations

The WIMSD-5B program is only capable of one-dimensional neutron transport calculations; therefore, modeling of the core cells is necessary. Cell modeling is used to calculate the generation of group constants. Group constant generation is intended to obtain the average value of the group constants in a cell by homogenizing the cell. Group constant calculations are performed for all core constituent materials under the conditions described above.

a. Fuel Model

The fuel calculation model is a multi-slab model. In this model, one fuel element, consisting of 21 fuel element plates, is made into 21 sequentially arranged slabs of material. The fuel element can be seen in Figure 2. Each slab consists of meat, cladding, and moderator with thicknesses of 0.027 cm, 0.038 cm, and 0.1925 cm, respectively [22, 23]. These slabs have lengths corresponding to the length of the active material. Other materials outside the active length are homogenized and normalized to the active length and are called the "extra region." In the fuel element, the extra region consists of AlMg2 and water, while the "meat" consists of ^{235}U and ^{238}U, the cladding consists of AlMg2, and the moderator consists of H_2O.

b. Control Element Model

The calculation model for the control material is divided into two regions. The first region is the active region, consisting of 15 fuel element plates arranged in 15 sequentially arranged slabs. The control element can be seen in Figure 3. The modeling is the same as for the fuel element, differing only in the extra region. The second region is the absorber region. This region is divided into 9 slabs and 1 extra region. The control rods in the neutron absorber region contain AgInCd and SS-321. To obtain a macroscopic cross-section of this region, it is combined with the first region in the calculations.

c. Non-fissionable Materials

For non-fissionable materials, the model is similar to that for fissionable materials, with a small amount of 235U added to the active slab and then homogenized throughout.

The general steps in reactor calculations are shown in Figure 4. Once the reactor shape is known, the process of preparing cross-sectional data is a crucial step. Macroscopic cross-sectional data can be obtained by generating cross-sectional images of the reactor core material using the WIMSD-5B software package. This program has been widely used for cross-sectional generation, particularly in Europe. This paper requires cross-sectional data for a silicide-fueled RSG-RAS core with a payload of 250 g. The cross-sectional data preparation uses a plate-type fuel modeled with a multi-slab, as shown in Figure 3. The fuel composition is U3Si_2-Al with an enrichment of 19.75%, calculated based on its elemental composition. Other important data can be seen in Table 1. This multi-slab model and fuel composition are used as input to the WIMSD-5B program. The fuel and coolant regions are homogenized to obtain a homogeneous lattice.

The direct scattering neutron (DSN) method was used to calculate the lattice parameters, dividing the neutron energy into four groups and using 138 mesh points. The cell geometry consists of four materials: fuel, cladding, coolant, and extra regions. The cell geometry was generated as a function of 17 burn-up degrees. The lattice parameters were calculated under the following conditions [27]:

- Hot, Xe, and Sm equilibrium (IFS4).
- Hot, Xe-free, and Sm equilibrium (IFS3).
- Cold, Xe-free, and Sm equilibrium (IFS2).
- Cold, Xe-free, and Sm equilibrium (IFS1).

3.2. Core Calculations

Core calculations were performed to achieve the equilibrium core of the RSG-GAS reactor. Core calculations were performed using the Batan-FUEL program. The flowchart for core calculations using the Batan-FUEL program is shown in Figure 2. The core is modeled in a two-dimensional X-Y geometry. In this calculation, the reactor power was set at 15 MW according to the experiment, and the effect of fuel loading on the RSG-GAS core on excess reactivity was determined. Fuel loading was carried out in stages, leaving 12% of fuel outside the core. Then the reactor was operated, most likely not yet critical but already approaching critical. Then, one by one, fuel was added to the core to obtain data on the amount of fuel in the core at the first critical point. The purpose of obtaining excess reactivity was so that the reactor could operate throughout its entire cycle. Because cycle length is a parameter that affects core reactivity, the core calculation was carried out for a cycle length that met the safety criteria, namely the criterion for the largest control rod stuck in one reactor still being sub-critical. Called a stuck rod, if the control rod with the highest reactivity value does not successfully enter the core, the reactor core must be in a sub-critical condition. If the stuck control rod criterion is not met, then the fuel position is rearranged. Likewise, for the peak power factor, if the maximum PPF value exceeds the specified limit, a reconfiguration is carried out where the position and location of the fuel are rearranged in the core so that the maximum PPF value is reduced. Core calculations were performed for 5 equilibrium tears, namely cores 95, 96, 97, 98, and 99. The RSG-GAS equilibrium core reactivity at BOC is an important condition in reactor core operation. The RSG-GAS equilibrium core reactivity at BOC conditions is a limitation in relation to the analysis of the control rod's ability to shut down the reactor. The determination of the equilibrium core reactivity value was carried out at the beginning of core operation because, conservatively, the equilibrium core reactivity value will ensure the reactor can operate safely throughout one cycle, namely 630 MWD. This core reactivity value will decrease during operation due to the burn-up factor so that the reactivity value decreases more compared to the excess reactivity value at the end of the equilibrium core due to the effect of the reduced amount of uranium in the reactor core.

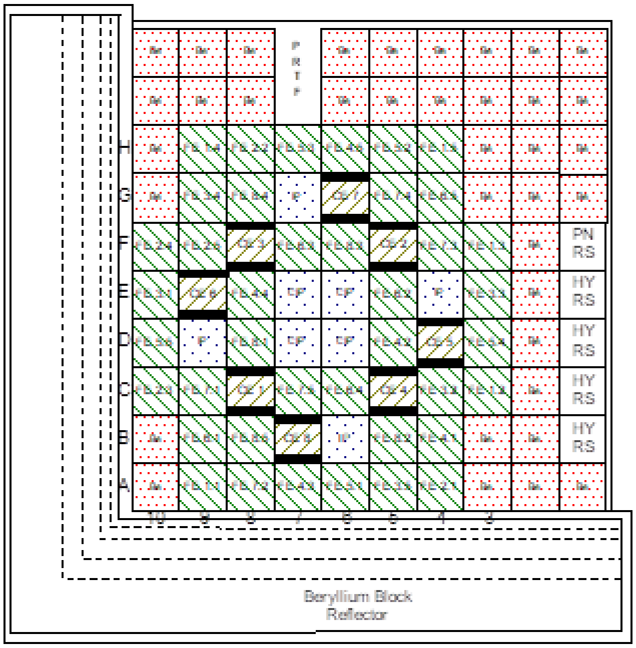


Figure 1 Core configuration of RSG-GAS research reactor [24]

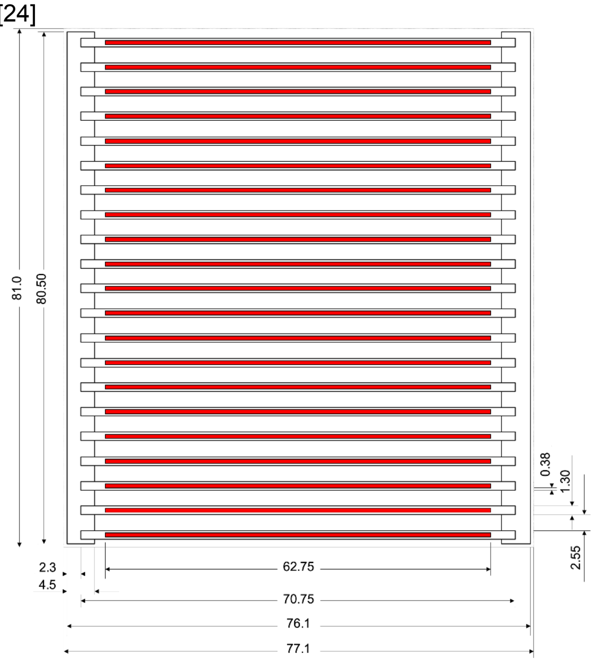


Figure 2 RSG-GAS Standard fuel element [25, 26]

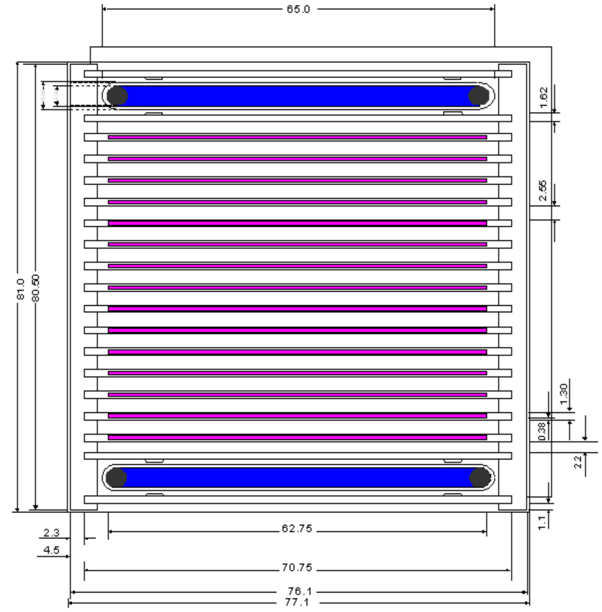


Figure 3 RSG-GAS standard control element [28]

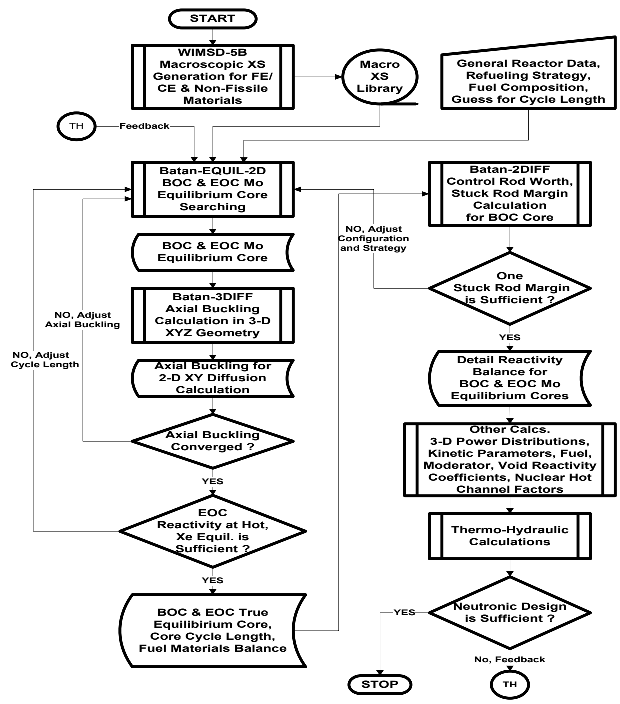


Figure 4 Reactor calculation for RSG-GAS core [29, 30]

4.0. RESULTS AND DISCUSSIONS

Table 1 shows the result of the calculation for the atomic density of uranium silicide. The atomic density of core material is used as data input for WIMS calculation to get the macroscopic cross-section.

Table 1 Atomic densities of the RSG-GAS fuel element.

Component	Nuclide	The density of an atom (Atom/b*cm)
Meat of fuel	^{235}U	1.50025×10^{-3}
	^{238}U	6.01895×10^{-3}
	^{29}Si	5.01267×10^{-3}
	^{27}Al	4.30311×10^{-2}
Cladding	^{24}Mg	1.36127×10^{-3}
	^{29}Si	1.72395×10^{-4}
	^{63}Cu	1.26989×10^{-5}
	^{55}Mn	8.81320×10^{-5}
	^{56}Fe	1.15597×10^{-4}
	^{52}Cr	9.31187×10^{-5}
	^{48}Ti	3.37079×10^{-5}
	^{27}Al	5.76030×10^{-2}
Coolant	^{1}H	6.65620×10^{-2}
	^{16}O	3.32810×10^{-2}
Extra region	^{24}Mg	5.60411×10^{-4}
	^{29}Si	3.34402×10^{-4}
	^{63}Cu	9.36022×10^{-5}
	^{55}Mn	1.21957×10^{-4}
	^{56}Fe	9.56977×10^{-5}
	^{52}Cr	4.12492×10^{-5}
	^{48}Ti	2.35808×10^{-5}
	^{27}Al	4.01204×10^{-2}
	^{16}O	2.02507×10^{-2}
	^{1}H	1.01253×10^{-2}

Table 2 shows the values of kinf as a function of burnup for WIMS calculation. The cross-section is dependent on the burnup of fuel. As greater burn-up, the atomic density is lower because the atomic density itself shows the amount of uranium in the fuel.

Table 2 k-inf of WIMSD-5B for fuel assembly calculations.

Steps	Atomic density (Atom/b-cm x 10^{-3})	Burn-up (%)	k-inf
1	1.5003	0.0	1.6027
2	1.4988	0.1	9
3	1.4913	0.6	1.5979
4	1.4240	5.0	7
5	1.3345	11.0	1.5387
6	1.2447	17.0	8
7	1.1543	23.0	1.5061
8	1.0648	29.0	4
9	0.9752	35.0	1.4823
10	0.8848	41.0	8
11	0.7944	47.0	1.4574
12	0.7037	53.0	8
13	0.5983	60.0	1.4303
14	0.4797	68.0	5
15	0.3753	75.0	1.4009
16	0.2702	82.0	9
17	0.0149	90.0	1.3686
			4
			1.3325
			9
			1.2924
			2
			1.2471
			5

1.1862
4
1.1035
2
1.0124
9
0.8948
8
0.7044
9

The cross-section form from the WIMS calculation was used as input for the Batan FUEL calculation. Figure 5 shows the results of the Batan-2DIFF calculation, which shows a k_{eff} value of 1, indicating that the initial critical reactor is at step 3. This means that the RSG-GAS core is critical with 31 fuel units and 8 control rods for the five cores (T-95 to 99). To increase reactivity, allowing the reactor to operate for one cycle of 42 days at 15 MW, fuel management calculations were performed, resulting in the addition of 9 more fuel units, for a total of 40 fuel units and 8 control rods. The calculation shows that the amount of ^{235}U in the core when fully charged is (250 gU x 40 FE) + (8 CE x 178.6 gU) = 11.4288 kg. However, to achieve the initial core configuration, not all of the fuel is fresh; some has already been burned, so the amount of uranium in the core should be less than the above value. Based on the results of Batan-2DIFF calculations, it can be seen in Table 2. From Table 3, it can also be seen that during one operating cycle of the 95th core of the RSG-GAS reactor, the ^{235}U burned was 0.737 kg for 42 days with a power of 15 MW. Likewise, the average 235U burned was almost the same for the 96-99th core.

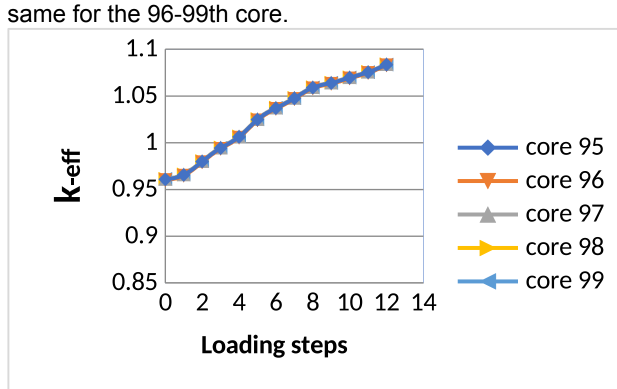


Figure 5: The result of Batan-FUEL calculation

4.1. Calculation of RSG GAS Equilibrium Core Reactivity

Table 3 shows that the RSG-GAS core is initially critical after the introduction of 39 fuels, or in step 3. After obtaining the keff value from the RSG-GAS initial equilibrium core modeling, the reactivity value must be calculated using the formula below:

$\rho = (k_{eff} - 1)/k_{eff}$ ....(1)

Based on equation 1 above, by entering the k_{eff} value from the initial equilibrium core modeling of RSG-GAS (T-95) of 1.0847735404, the reactivity value (ρ) is 7.815%, while the experimental result is 7.049%. The deviation between the calculated and experimental results is approximately 10.86%. Table 4 shows the calculated and experimental results for the RSG-GAS cores from the T-95 to T-99 configurations. The table shows that the average deviation for each core is 10.4%. This result is quite good considering the reactor has been operating for over 35 years, and the calculated results have consistently exceeded the experimental results.

The equilibrium core reactivity value (BOC) in the RSG-GAS LAK is 9.2%. This value is based on a design where a single-cycle reactor operates at 750 MWD, meaning the reactor is operated at 30 MW for 25 days. However, currently the reactor is operated at 15 MW for 42 days (630 MWD), so the higher reactivity must also be adjusted to ensure fuel management continues according to the program. The relative difference is still in 10%, indicating that the verification calculation values between the experiment and the calculation are still consistent.

Table 3: excess reactivity of RSG-GAS core from experiment and calculation.

No	T-95	T-96	T-97	T-98	T-99
0	0.9620980024	0.9623998404	0.9625760913	0.9629567862	0.9632625580
1	0.9668713808	0.9671828151	0.9673386216	0.9677312374	0.9680194855
2	0.9812222719	0.9815452099	0.9816059470	0.9820129871	0.9822196960
3	0.9957221150	0.9960706830	0.9960802794	0.9964949489	0.9966970086
4	1.0076000690	1.0079315901	1.0079684258	1.0083690882	1.0085544586
5	1.0260281563	1.0263584852	1.0491776466	1.0267719030	1.0269358158
6	1.0383454561	1.0386819839	1.0605804920	1.0390343666	1.0391842127
7	1.0487916470	1.0490999222	1.0491776466	1.0495198965	1.0496627092
8	1.0601820946	1.0604447126	1.0605804920	1.0609076023	1.0610564947
9	1.0650517941	1.0653258562	1.0654425621	1.0657695532	1.0659073591
10	1.0708836317	1.0711554289	1.0712666512	1.0716007948	1.0717403889
11	1.0766471624	1.0769115686	1.0770297050	1.0773600340	1.0775005817
12	1.0847735405	1.0850698948	1.0851515532	1.0854682922	1.0856142044
Excess reactivity C	7,815%	7,840%	7,847%	7,874%	7,886 %
Excess reactivity E	7,049%	7,218%	7,109%	7,009%	7,168%
D = (C-E) x100 %/E	10,86%	8.61%	10,38%	12.33%	10,01%

*) C= Calculation, E= Experiment, D= Deviasion

Table 4 shows the result of the calculation for the stuck rod condition in the RSG-GAS core after 35 years of operation. The stuck rod condition is still fulfilling the safety operation; that said, the reactor must be sub-critical as one of the biggest reactivity worth being stuck in the core. From this table, the RSG-Gas reactor is still sub-critical on the stuck rod condition for T95-T99.

Table 4. Stuck-rod conditions of the RSG-GAS core at the Beginning of the Cycle

No.	Condition		T-95	T-96	T-97	T-98	T-99
1	All up	-	1.0847735405	1.0850698948	1.0851515532	1.0854682922	1.0856142044
2	All dn	-	0.9557388425	0.9559066892	0.9560167193	0.9562630653	0.9563977122
3	7 down	G-6 up	0.9785479903	0.9786638618	0.9789087176	0.9791463614	0.9793541431
4	7 down	F-8 up	0.9828602672	0.9830501080	0.9831339717	0.9833763838	0.9834898710
5	7 down	F-5 up	0.9771115780	0.9771984816	0.9774227738	0.9776748419	0.9778996110
6	7 down	E-9 up	0.9810011387	0.9812525511	0.9812685251	0.9815126657	0.9815921187
7	7 down	D-4 up	0.9790550470	0.9792044759	0.9794143438	0.9796509743	0.9815921187

8	7 down	C-8 up	0.9759805799	0.9762070775	0.9762018919	0.9764679670	0.9765787721
9	7 down	C-5 up	0.9805204272	0.9807587862	0.9808672667	0.9811515808	0.9812579155
10	7 down	B-7 up	0.9780707359	0.9783296585	0.9783245325	0.9786506891	0.9787822962

From Table 4, it can be seen that the shutdown margin value can be calculated based on the position of the control rods, all of which are below, with Equation 1. According to SAR, each reactor core must provide a negative reactivity called a shutdown margin sufficient to shut down the reactor in an emergency. The shutdown margin reactivity value is derived from the results in Table 5. Table 6 shows some neutronic parameters from experimental results, including the shutdown margin. The results show that the calculation value from the WIMS/Batan FUEL code differs little after more than 35 years of operation compared to the experimental results. This indicates that the RSG-GAS reactor is still feasible to operate, irradiate samples, and produce radioisotopes for research.

Table 5 parameter neutrik teras reaktor RSG-GAS (T95-T99)

Core number	CR pos BOC (cold, clean)	CR pos EOC (hot, poisson)	Energy per cycle (MWD))	Fuel loading (FE/CE)	ρ CR Total (%)	ρ Core excess (%)	ρ Shut down (%)	ρ CR max (%)	ρ Shut down margin (%)
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
95	280/280	548/537	625.64	5/1	-12.92	+ 7.04	-5.88	- 1.86	- 4.02
96	276/276	555/553	625.01	5/1	-13.02	+ 7.21	-5.81	- 1.85	- 3.96
97	275/275	511/511	625.00	5/1	-12.91	+ 7.10	-5.81	- 1.84	- 3.97
98	286/286	541/546	625.01	5/1	-13.20	+ 7.00	-6.20	- 1.85	- 4.35
99	284/284	546/547	625.08	5/1	-13.27	+ 7.16	-6.12	- 1.90	- 4.22

Table 6 shows the average deviation of the shutdown margin of the neutronic parameter in the RGS-GAS reactor between calculation and experimental results after more than 35 years of operation. The different come from the actual burn up in fuel, they are difficult to estimate when calculating the result.

Table 6 Comparison between calculation and experiment results of shutdown margin for RSG-GSAS cores

Core	Shutdown margin from experiment (%)	Shutdown margin from calculation (%)	Deviation (C-E *100%/E)	Remark
T-95	-4.02	-4.631	15.19%	Average of deviation is 12.22%
T-96	-3.96	-4.613	16.49%	
T-97	-3.97	-4.601	15,89%	
T-98	-4.35	-4.574	5,15%	
T-99	-4.22	-4.559	8,38%	

Calculating the thermal neutron flux in the RSG-GAS core irradiation facility is the ultimate goal of reactor utilization. The thermal flux in the D-6, D-7, E-6, and E-7 irradiation areas is crucial because these are the irradiation facilities in the center of the core, where the highest thermal neutron flux is expected. Table 8 shows that the thermal neutron flux in the center of the core irradiation facility is highly dependent on the amount of fuel in the core. The greater the amount of fuel, the lower the neutron flux. This means that the amount of uranium in the core determines the thermal neutron flux in the irradiation facility. The greater the amount of uranium in the core, the lower the thermal neutron flux. This is inversely proportional to the operating cycle length. The greater the amount of uranium in the core, the longer the cycle length, so the optimal value is used in the design of the research reactor core.

Table 7. Result of calculation for thermal neutron fluxes at the irradiation position ($x10^{14}$ n/cm^2s)

Steps	E-7	E-6	D-7	D-6	D-9	B-6	E-4	G-7	Loading
0	2.2363	2.2342	2.2196	2.2142	1.6647	1.6081	1.5289	1.9567	0
1	2.1586	2.1552	2.1538	2.1494	1.6074	1.8464	1.4607	1.8105	A-4
2	2.0291	2.0118	2.0399	2.0191	1.6882	1.9979	1.2196	1.5203	A-9
3	1.9541	1.9473	2.0008	1.9894	1.5711	1.9856	1.1579	1.3115	B-5
4	1.8668	1.8678	1.9174	1.9172	1.4016	1.9793	1.2688	1.1805	C-3
5	1.8072	1.8269	1.8722	1.8916	1.2338	2.0297	1.4055	1.0356	C-4
6	1.7463	1.7475	1.8148	1.8155	1.4899	1.9151	1.2394	0.9911	C-10

7	1.7058	1.7214	1.7692	1.7838	13382	1.8566	1.4215	0.9419	D-3
8	1.6502	. 1.6859	1.6899	1.7231	1.1764	1.6791	1.6246	0.9587	F-3
9	1.6326	1.6517	1.6639	1.6814	1.2831	1.5906	1.5238	1.0241	F-10
10	1.6299	1,6385	1.6424	1.6506	1.2968	1.4875	14651	0.9938	G-8
11	1.6022	1.6055	1.6056	1.6088	1.2854	1.3792	1.4024	1.1253	H-8
12	1.5544	1.5541	1.5539	1.5537	1.2854	1.2494	1.3005	1.2404	H-9

The calculation results from the Batan-2DIFF program in Table 7 are in accordance with expectations: in the center area, the average thermal neutron flux is 1.5 x10^{14} n/cm^{2}s at a power of 15 MW. In the irradiation position areas D-9, B-6, E4, and G-7, it is even smaller, namely an average of around 1.2 x10^{14} n/cm^{2}s. These calculation results are in accordance with expectations, where IP is used for topaz irradiation while CIP is used for Mo and other radioisotopes irradiation.

K	BS+59	B-29	B-30	PRTF	B-20	B-13	B-8	BS+10	B-5	B-2
J	B-28	BS+58	B-22	PRTF	B-21	B-23	B-24	B-4	BS+52	B-15
H	B-26	1.0161 1.0161 1.0156 1.0137 1.0152	0.9603 0.9588 0.9590 0.9577 0.9573	0.9176 0.9177 0.9210 0.9210 0.9205	0.9305 0.9286 0.9342 0.9328 0.9331	0.8372 0.8366 0.8407 0.8414 0.8412	0.9880 0.9849 0.9885 0.9865 0.9859	B-19	B-17	BS+51
G	B-16	1.0571 1.0565 1.0546 1.0543 1.0537	0.9030 0.9014 0.8996 0.9002 0.8995	AL-4	0.9727 0.9733 0.9753 0.9751 0.9776	0.8577 0.8549 0.8575 0.8566 0.8630	0.8782 0.8769 0.8780 0.8772 0.8825	B-40	BS+57	B-14
F	1.1164 1.1157 1.1145 1.1136 1.1115	1.1618 1.1619 1.1594 1.1595 1.1576	1.1838 1.1846 1.1834 1.1832 1.1830	1.1200 1.1231 1.1242 1.1274 1.1282	0.9521 0.9524 0.9466 0.9490 0.9508	1.1971 1.1966 1.1991 1.1985 1.1984	0.9211 0.9139 0.9183 0.9187 0.9227	1.0529 1.0509 1.0524 1.0518 1.0504	B-32	PNRA
E	1.0536 1.0607 1.0580 1.0549 1.0537	1.0358 1.0369 .1.0351 1.0367 1.0361	1.1887 1.1959 1.1931 1.1914 1.1916	AL-6	AL-3	1.1268 1.1247 1.1269 1.1299 1.1322	AL-8	1.0426 1.0411 1.0418 1.0413 1.0408	B-34	HYRA
D	0.9962 0.9959 0.9935 0.9979 0.9961	AL-2	1.0373 1.0410 1.0364 1.0358 1.0348	AL-5	AL-7	1.1752 1.1779 1.1795 1.1777 1.1776	1.0878 1.0866 1.0893 1.0891 1.0893	0.8964 0.8948 0.9013 0.8998 0.8987	B-36	HYRA
C	1.0278 1.0276 1.0259 1.0241 1.0236	0.9179 0.9179 0.9140 0.9135 0.9107	1.2252 1.2269 1.2254 1.2249 1.2210	1.0035 1.0011 1.0000 0.9987 0.9980	1.0940 1.0960 1.0947 1.1015 1.1000	1.1106 1.1120 1.1127 1.1124 1.1127	1.0924 1.0931 1.0924 1.0920 1.0909	1.0872 1.0864 1.0867 1.0855 1.0837	B-37	HYRA
B	BS+54 NS	0.8524 0.8518 0.8506 0.8485 0.8543	0.8110 0.8155 0.8116 0.8119 0.8102	0.9033 0.9044 0.9050 0.9060 0.9058	AL-1	0.9025 0.9038 0.9009 0.8999 0.9007	0.9875 0.9852 0.9921 0.9912 0.9888	B-06	B-11	HYRA
A	B-10	0.9495 0.9477 0.9464 0.9471 0.9444	0.7464 0.7438 0.7423 0.7434 0.7457	0.9551 0.9580 0.9576 0.9564 0.9562	0.9676 0.9669 0.9645 0.9711 0.9700	0.9425 0.9466 0.9462 0.9454 0.9457	0.9642 0.9631 0.9614 0.9611 0.9615	B-03	BS+56	B-1
	10	9	8	7	6	5	4	3	2	1

Figure 6 Calculated radial PPF values for T95-T99 at BOC from Batan-FUEL code

The power peaking factor values are also an important parameter for core analysis. Figure 6 shows the radial PPF values for T95 - T99 cores at BOC. The PPF values are consistent across all cores; the highest value is in the C-8 position, but the highest value differs in every core. For core T95, T-96, T97, T98, and T99 are 1.2250, 1.2269, 1.2249, 1.2210, respectively. They are still below the value that is allowed for the safety limit. The RSG-GAS core is still consistent with the safety limit operation even though the reactor has been operated for more than 35 years.
The radial ppf values at BOC for T-95 and T-99 are lower than the maximum limit, namely 1.4 (SAR)

4.0. CONCLUSION

The verification calculations using the WIMSD-5B/Batan-FUEL computer program revealed that the BOC equilibrium overcore reactivity value was in line with expectations, with a 10% difference between the calculated and experimental values. This result is satisfactory and can be used for periodic safety parameter reviews. The calculated RSG-GAS core neutronic parameters are appropriate, and none violate the reactor's operational safety limits. The safety limits include the stuck rod condition, shut-down margin, and overcore reactivity, which must be within the control rod reactivity values.

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References

[1] T. SURBAKTI, S. PINEM, T. M. SEMBIRING, “Analisis Pengaruh Densitas Bahan Bakar Silisida Terhadap Parameter Kinetik Teras Reaktor RSG-GAS”, Jurnal Penelitian Fisika dan Aplikasinya. Vol. 3 no. 1, pp. 19-30. 2013.

[2] S. PINEM, S. JATI, T. SURBAKTI, , T, M. SEMBIRING, “Optimization of Radioisotope Production at RSG-GAS Reactor Using Deterministic Method”, Teknologi Indonesia, Vol. 35, pp. 37–45, 2016.

[3] S. Pinem, Sembiring T.M., Surbakti T. Core conversion design study of TRIGA Mark 2000 Bandung using MTR plate type fuel element. Int. J. Nucl. Energy Sci. Technol. Vol. 12, no. 3, pp. 222–238. 2018.

[4] S. PINEM, P. H. LIEM, T. M. SEMBIRING, T. SURBAKTI, “Fuel element burn-up measurements for the equilibrium LEU silicide RSG GAS (MPR-30) core under a new fuel management strategy”, Ann. Nucl. Energy, Vol. 98, 2016.

[5] L. SUPARLINA L, T. SURBAKTI, “Analisis Pola Manajemen Bahan Bakar Desain Teras Reaktor Riset Tipe MTR”, Jurnal Tri Dasa Mega. Vol. 14. no. 3, pp. 89-99. 2014.

[6] T. SURBAKTI T, S. PINEM, T. M. SEMBIRING, L. SUPARLINA, J. SUSILO, “Desain Konseptual Teras Reaktor Riset Inovatif Berbahan Bakar Uranium-molibdenum dari Aspek Neutronik”, Jurnal Tri Dasa Mega. Vol. 3 no.14, pp. 178-191. 2012.

[7] T SURBAKTI, P PURWADI, “Karakteristik reaktivitas teras kerja RSG-GAS selama 30 tahun beroperasi”, Jurnal Penelitian Fisika dan Aplikasinya (JPFA). Vol. 7 no.1, pp. 13-26. 2017.

[8] P. H. LIEM, T SURBAKTI, D HARTANTO, “Kinetics parameters evaluation on the first core of the RSG GAS (MPR-30) using continuous energy Monte Carlo method”, Progress in Nuclear Energy. Vol. 109, pp. 196-203. 2018.

[9] T SURBAKTI, M IMRON, “Fuel burn-up calculation for working core of the RSG-GAS research reactor at BATAN Serpong”, Jurnal Penelitian Fisika dan Aplikasinya (JPFA). Vol. 7, no. 2, pp. 89-101. 2017.

[10] T SURBAKTI, P PURWADI, “Analysis of Neutronic Safety Parameters of the Multi-Purpose Reactor–Gerrit Augustinus Siwabessy (RSG-GAS) Research Reactor at Serpong”, Jurnal Penelitian Fisika dan Aplikasinya (JPFA). Vol. 9, no. 1, pp. 78-91. 2019.

[11] SURBAKTI T., PINEM S., SEMBIRING T.M., SUBEKTI M., SUNARYO G.R. Preliminary study for alternative conceptual core design of the MTR research reactor. J. Phys. Conf. Ser. 2018. 962(1)

[12] T. Surbakti, Pinem S., Sembiring T.M., Hamzah A., Nabeshima K. Calculation of control rods reactivity worth of RSG-GAS first core using deterministic and Monte Carlo methods. Atom Indones. Vol. 45, no. 2, pp. 69–79. 2019.

[13] Lily Suparlina, Tukiran Surbakti, Analisis Pola Manajemen Bahan Bakar Reaktor Riset Tipe MTR. Jurnal Teknologi Reaktor Nuklir Tri Dasa Mega 16 (. 2), 89-99

[4] Tukiran Surbakti, Surian Pinem, Lily S, Farisy Y. Evaluation of the New ENDF/B-VIII.0 Nuclear Data Library on Neutronic Parameters Calculations for First Core RSG-GAS Reactor. 2026, Nuclear Technology.

[15. Surian Pinem, Tukiran Surbakti, Endiah Puji Hastuti, Wahid Luthfi. Transition Core from oxide to silicide fuel for Indonesian multipurpose research reactor RSG-GAS. 2025, Nuclear Technology.

[16] S. Pinem, FY. Sulistyo, W. Luthfi, L. Suparlina, T. Surbakti, A. Rohanda. Reactor Safety Analysis of Electroplated LEU Target Irradiation for 99Mo Production in the RSG-GAS Reactor. 2025, Nuclear Science and Engineering.

[17] .Liem P. H., 1994. Development and Verification of BATAN'S Standard, Two-Dimensional Multigroup Neutron Diffusion Codes (BATAN-2DIFF), Atom Indonesia, 20 (2), 1-19

[18] Askew, J. R., Fayers, F. J., Kemshall, P .B., 1966. A general description of the lattice code WIMS. J. Brit. I. Energy Soc. 5 (4), 564.

[19] Peng Hong Liem 1996. Batan-FUEL: a general in-core fuel management code. Atom Indonesia vol 22 (2), 67-80.

[20] Liu, S. C. et al., 2015. Neutronics comparative analysis of plate-type research reactor using deterministic and stochastic methods. Ann. Nucl. Energy 79, 133–142.

[21] Pinem S., et al. 2016 Fuel element burn-up measurements new fuel management strategy. Annals of Nuclear Energy 98, 211-217.

[22 Aslina Br Ginting, Peng Hong Liem, 2015. Absolute burnup measurement of LEU silicide fuel plate irradiated in the RSG-GAS multipurpose reactor by destructive radiochemical technique. Annals of Nuclear Energy 85, 613-620.

[23] Pinem S., Sembiring T.M., Liem P.H., Neutronic and Thermal-Hydraulic Safety Analysis for the Optimization of the Uranium Foil Target in the RSG-GAS Reactor., Atom Indonesia, 42 (3), 123-128.

[24] Liem P. H., 1997. Development Of An In-Core fuel Management Code For Searching Equilibrium Core in 2-D Reactor Geometry (BATAN-EQUIL-2D) Atom Indonesia 23 (1), 1-19.

[25] Liem P. H., Tagor M. S., 2010. Design of transition cores of RSG GAS (MPR-30) with higher loading silicide fuel. Nuclear Engineering and Design 240 (6), 1433-1442.

[26] Liem P.H., et al. 2013. Nondestructive burnup verification by gamma-ray spectroscopy of LEU silicide fuel plates irradiated in the RSG GAS multipurpose reactor. Annals of Nuclear Energy. 56, 57-65.

[27]. Halsall, M. J., 1997. WIMSD, A Neutronic Code for Standard Lattice Physics Analysis, Distributed by the NEA Databank, NEA-1507/04. Research Reactors, 2010. Purpose and Future. IAEA, Vienna.

[28] Roth, M. J., Macdougall, J.D., Kemshell, P.B., 1967. The Preparation of Input Data for WIMS, AEEW-R538.

[29]. Hosoya, T., Kato, T., Murayama, Y., 2007. Investigation of JRR-3 control rod worth changed with burn-up of follower fuel elements. In: International Research Reactors: Safe Management and Effective Utilization, Australia, IAEA, 2010.

[30] Askew, J. R., Fayers, F. J., Kemshall, P . B., 1966. A general description of the lattice code WIMS. J. Brit. I. Energy Soc. 5 (4), 564.